UPROOTED / FLL SENSOR STATION

Arduino C++ quick reference • Heltec V4 • Kit v0.2.3

PAGE 1 / ARDUINO CODE BASICS

Arduino sketches use C++. These are the pieces of syntax you will see when completing the five TODO functions. Capital letters and punctuation matter!

1 Two functions run your program

```
void setup() {        // runs ONCE when the board starts
  station.beginBoard();
  setupPT100();
  setupLoRa();
  setupAP();
}
void loop() {         // repeats while the board is on
  station.update();
  readTemp();
  broadcastTemp();
}
```

Your starter sketch already has these functions and calls. Edit the TODOs instead of replacing setup() or loop().

2 Decode the symbols

;	Ends a statement: station.update();
{ }	Groups the code inside a function or if statement.
( )	Holds inputs for a function: station.setupAP(...);
//	Adds a comment; the board ignores the rest of that line.
"text"	A string: Serial.println("LoRa ready");

3 Variables store information; if checks a condition

```
bool radioReady = station.setupLoRa();  // true / false
if (radioReady) {
  Serial.println("LoRa ready");
}
Reading r = station.local();
if (r.valid) Serial.println(r.temperatureC);
```

bool = yes/no • float = decimal number • unsigned long = large nonnegative number

4 Time your radio without freezing the loop

```
unsigned long now = millis();  // milliseconds since boot
if (now - lastSent >= BROADCAST_MS) {
  if (station.broadcastTemp()) lastSent = now;
}
```

The starter sketch already has a broadcast timer. Keep its retry checks. Do not use delay(12000): that would pause the OLED, Wi-Fi page, and LoRa receiver.

READY? Review the terms on page 2, then open the starter sketch and begin building on page 3. Edit the TODO functions, not the whole file.

UPROOTED / FLL SENSOR STATION

Quick-reference terms • Heltec V4 • Kit v0.2.3

PAGE 2 / WORDS YOU WILL USE

Use this page when you see an unfamiliar word in the sketch, website, or Serial Monitor.

HARDWARE & SENSOR WORDS

ESP32-S3	The tiny programmable computer on our Heltec V4. It runs the code, Wi-Fi, and sensor connections.
Heltec V4	Our circuit board: it combines an ESP32-S3, an OLED screen, and a LoRa radio.
PT100 / RTD	A platinum temperature sensor. A PT100 is about 100 ohms at 0°C; its resistance changes with temperature.
MAX31865	A chip that measures the PT100 resistance, sends the result to the ESP32, and reports wiring faults.
SPI	A fast, short wired connection between chips using clock, data, and chip-select signals.
GPIO	A programmable pin on the ESP32 used to connect sensors and other electronics.
OLED	The small onboard screen that shows temperature and connection status.

RADIO & NETWORK WORDS

LoRa radio	A long-range, low-power wireless link for sending small messages. It is different from Wi-Fi.
Packet	One short message sent over the radio, such as a node ID and a temperature reading.
Node	One Heltec board participating in the sensor network.
Relay	When a node receives a packet and passes it along so another board can hear it.
Mesh network	A network where nodes can route messages through other nodes, often using alternate paths.
Exit node	In THIS project, the board that shows LoRa sensor readings on a local Wi-Fi webpage; not an internet exit.
Wi-Fi AP	Access point: a board-created Wi-Fi network you connect your phone or laptop to.
Captive portal	The webpage that a phone may offer to open after joining the board's Wi-Fi.
RSSI	Received signal strength. A more negative number (like -110 dBm) usually means a weaker signal.

IMPORTANT This project uses raw LoRa packets and one-hop relaying. It is NOT LoRaWAN or a fully routed, self-healing mesh network.

UPROOTED / FLL SENSOR STATION

Heltec WiFi LoRa 32 V4 • MAX31865 + 2-wire PT100 • Kit v0.2.3

PAGE 3 / START THE HARDWARE

01 Open the starter project

Install the Heltec board support and HeltecPT100Kit v0.2.3. Open Uprooted_Sensor_Station.ino in Arduino IDE; keep website.h beside it. Connect the PT100 and attach the LoRa antenna.

The sketch already contains your five functions and the station object. Replace the TODO comments with the code below. You do not need to create a station name.

02 Complete setupPT100()

Start the MAX31865. If it fails, keep going: the board can still act as an exit node for other sensors.

```
if (station.setupPT100()) {
  Serial.println("PT100 ready");
} else {
  Serial.println("Exit node: no MAX31865");
}
```

03 Complete setupLoRa()

Start the radio. A successful setup should print “LoRa ready” in Serial Monitor.

```
if (station.setupLoRa()) {
  Serial.println("LoRa ready");
} else {
  Serial.println("LoRa setup failed");
}
```

04 Complete setupAP()

Pass in your Wi-Fi password AND the HTML stored in website.h. Print the network name and IP address.

```
if (station.setupAP(WIFI_PASSWORD, STUDENT_HTML)) {
  Serial.println(station.wifiName());
  Serial.println(station.wifiIP());
} else {
  Serial.println("AP setup failed");
}
```

05 Check the startup order

The starter sketch already has this setup() sequence. Keep the calls in this order:

```
void setup() {
  station.beginBoard();
  setupPT100(); setupLoRa(); setupAP();
}
```

FIRST TEST ☐ Upload ☐ Open Serial Monitor (115200 baud) ☐ Look for PT100 / LoRa / AP startup messages

UPROOTED / FLL SENSOR STATION

Heltec WiFi LoRa 32 V4 • MAX31865 + 2-wire PT100 • Kit v0.2.3

PAGE 4 / READ, BROADCAST & TEST

06 Complete readTemp()

Inside readTemp(), keep station.readTemp() and the existing two-second print timer. Replace TODO 4 with:

```
Reading r = station.local();
if (r.valid) {
  Serial.print(r.temperatureC, 1);
  Serial.println(" C");
} else {
  Serial.print("Sensor fault: 0x");
  Serial.println(r.fault, HEX);
}
```

07 Complete broadcastTemp()

Keep the starter's millis() checks: 12 seconds between broadcasts, 1 second between retries. Replace TODO 5:

```
if (station.broadcastTemp()) lastSent = now;
```

Do NOT add delay(12000); station.update() must keep running for Wi-Fi, OLED, and incoming LoRa messages.

08 Create your Uprooted webpage

Keep website.h open next to your sketch. The full HTML build steps are on page 5. For now, try this simple page title and description:

```
<h1>Uprooted Sensor Station</h1>
<p>Live temperature from our PT100!</p>
```

Keep the existing IDs and <script> block in website.h. If the MAX31865 fails, the page automatically shows exit-node mode with remote sensors.

09 Upload and test the whole system

In loop(), keep station.update(); readTemp(); broadcastTemp();. Upload again after every website.h change. Connect to PT100-UPROOTED-XXXX Wi-Fi. The captive portal may open, or visit http://192.168.4.1.

SYSTEM CHECK / AFTER PAGE 5

☐ OLED shows local temperature OR EXIT NODE and the remote-sensor count

☐ Website shows local PT100 OR remote LoRa sensors in exit-node mode

☐ Serial Monitor shows temperature OR MAX31865 startup failure

☐ A second Heltec board appears under remote LoRa sensors

UPROOTED / FLL SENSOR STATION

Heltec WiFi LoRa 32 V4 • MAX31865 + 2-wire PT100 • Kit v0.2.3

PAGE 5 / BUILD YOUR WEBPAGE

10 Find the HTML

Open website.h in Arduino IDE. Edit the webpage between R"UPROOTEDHTML(and)UPROOTEDHTML";. Leave those two wrapper lines, the <script> section, and the closing tags in place.

11 Build the page content

Edit the headings and text in the existing <main>. Do not replace the entire <main> section: the exit-node and remote-sensor elements need to stay in place.

```
<h1>Uprooted Sensor Network</h1>
<!-- Change this description -->
<p>Our team's live LoRa temperature readings!</p>
<!-- Keep #modeCard, #localCard, #sensors,
    and the existing <script> block. -->
```

12 Change the look with CSS

Find the <style> section near the top of website.h. Change or add these rules to set the background and card colors. Try your own hex colors.

```
body { background: #11251b; color: white;
      font-family: Arial, sans-serif; padding: 24px; }
.card { background: #1c3d2b; padding: 18px;
       border-radius: 12px; margin: 12px 0; }
```

13 Upload and see live data

Save website.h and upload. Join PT100-UPROOTED-XXXX Wi-Fi. If the captive portal does not open, visit http://192.168.4.1. Try with and without a MAX31865.

HOW THE LIVE DATA WORKS

The library supplies /api/status and /api/nodes. The existing JavaScript shows a local PT100 when available; otherwise, EXIT NODE and cards for active LoRa sensors. Keep its IDs and <script> block unchanged.

FINAL CHECK ☐ Custom heading ☐ New colors ☐ PT100 mode ☐ Exit-node mode ☐ Remote LoRa sensors